Analyzing data on the level of cognitive processes - ESCoP 2025

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Solutions: Exercise 5 - EZ-Diffusion Model

In this exercise, we will have a look at estimating the parameters from the diffusion decision model using the hierarchical Bayesian implementation of the ez-Diffusion Model. This implementation allows for a fast parameter estimation while still providing acceptable recovery of three parameters of interest:

1. drift rate = the rate of evidence accumulation
2. boundary separation = the seperation of decision thresholds
3. non-decision time = the time for non-decisional processes.

Your task is to load one of the data sets we prepared for this exercise, either `exc4_VisualSearch.txt` or `exc5_ColorJudgement.txt`.

The *Color Judgement* data set contains data of 50 participants that completed a Color Judgement task. In this task participants saw an array of blue and red dots and should decide if the array contained more blue or red dots. There were two difficulty levels, the easy level in which 60% of the dots were either blue or red, and the difficult level in which 52% of the dots were either blue or red.

The *Visual Search* data set contains data for 33 participants that completed the Visual Search task. In this task, participants were presented a color cue, a positive (the target color), a negative (the non target color) or neutral (not part of the visual search display). Participants had to look for a "C" that either looked upwards or downwards and report it´s direction with the arrow key.

Solution

```
# Load required packages
library(bmm)
library(tidyverse)
library(here)
```

1) aggregate the data, so that we can estimate the `ezdm`

First, we need to aggregate the data so that we can analyze our data with the `ezdm` using the `bmm` function. For this, we need to calcualte the mean reaction time, the varianve of the reaction time, the number of correct responses, and the number of trials in each cell of the experimental design.

```
ezdm_data <- read.table(here("data","exc5_ColorJudgementTask.txt"),
                        header = T, sep = ",")

ezdm_data_agg <- ezdm_data %>%
  filter(rt > 150, rt < 5000) %>%
  summarise(
    mean_rt = mean(rt/1000),
    var_rt = var(rt/1000),
    n_correct = sum(correct),
    n_trials = n(),
```

```r
    .by = c("participant_id", "condition", "correct_response")
  )

ezdm_data_agg$condition <- as.factor(ezdm_data_agg$condition)
contrasts(ezdm_data_agg$condition) <- bayestestR::contr.equalprior
```

2) to specify the `ezdm` model object matching the aggregated data of the task

Next, we need to specify the model object to link which data columns are used to identify the `ezdm`

```r
ezdm_model <- ezdm(mean_rt = "mean_rt", var_rt = "var_rt", n_correct = "n_correct",
                   n_trials = "n_trials")
```

3) specify a model formula predicting which parameters you assume to vary as a function of the experimental manipulations (i.e. difficulty level or cue type)

Now, we can consider which parameters are affected by the experimental manipulations. First, let's see how the `ezdm` parameter vary as a function of the difficutly manipulation.

```r
ezdm_formula <- bmf(
  drift ~ 1 + condition + (1 + condition | participant_id),
  bound ~  1 + (1 | participant_id),
  ndt ~  1 + condition + (1 + condition | participant_id)
)
```

4) estimate the model using the `bmm` function and interpret the results with respect to the experimental effects on the model parameters

Finally, we can estimate the model using our aggregated data, the `bmmodel` object we created and the `bmmformula` specifying how the model parameters should change over experimental condition.

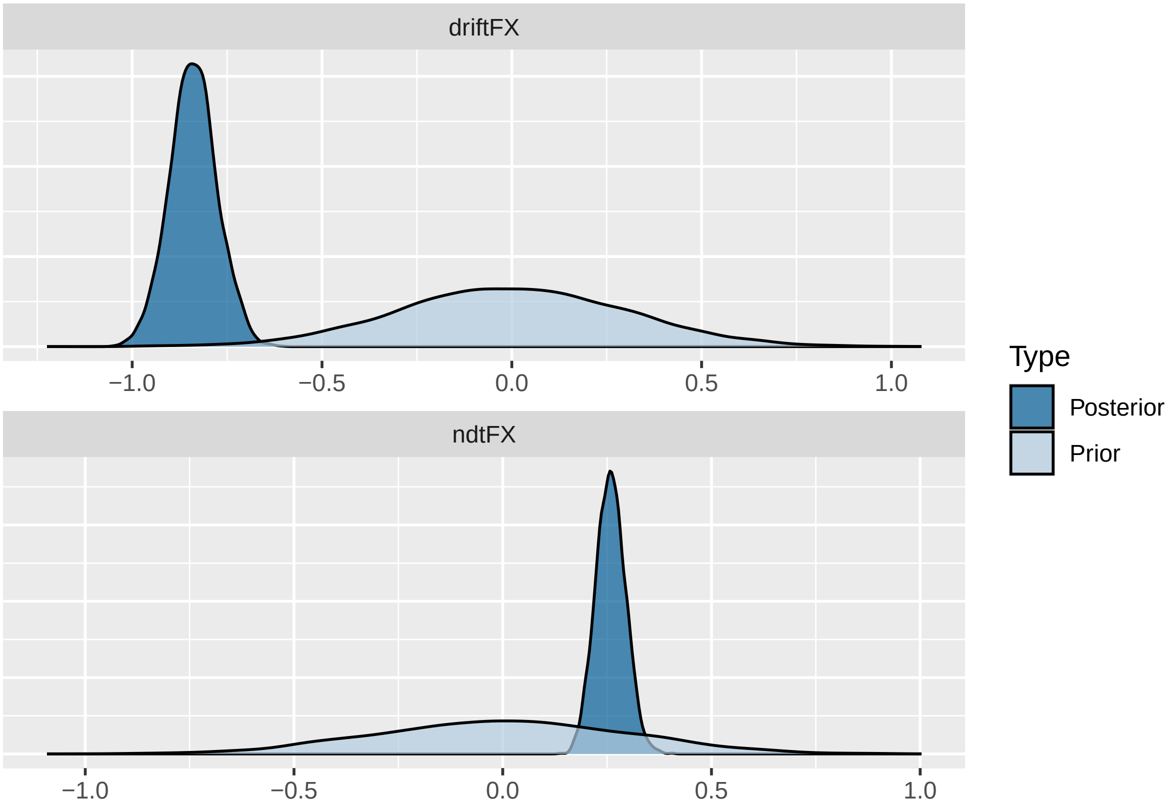
```r
ezdm_fit <- bmm(
  formula = ezdm_formula,
  data = ezdm_data_agg,
  model = ezdm_model,
  cores = 4,
  sample_prior = TRUE,
  refresh = 0,
  file = here("models","ezdm_colortask_exc5")
)
```

```r
summary(ezdm_fit)
```

```
## Loading required namespace: rstan

##   Model: ezdm(mean_rt = "mean_rt",
##               var_rt = "var_rt",
##               n_correct = "n_correct",
##               n_trials = "n_trials")
##   Links: drift = identity; bound = log; ndt = log
## Formula: drift ~ 1 + condition + (1 + condition | participant_id)
##          bound ~ 1 + (1 | participant_id)
##          ndt ~ 1 + condition + (1 + condition | participant_id)
##    Data: ezdm_data_agg (Number of observations: 200)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
```

```
##
## Multilevel Hyperparameters:
## ~participant_id (Number of levels: 50)
##                                         Estimate Est.Error l-95% CI u-95% CI Rhat
## sd(drift_Intercept)                         0.49      0.05     0.40     0.62 1.01
## sd(drift_condition1)                        0.44      0.05     0.35     0.54 1.01
## sd(bound_Intercept)                         0.27      0.03     0.22     0.34 1.00
## sd(ndt_Intercept)                           0.28      0.04     0.20     0.37 1.00
## sd(ndt_condition1)                          0.16      0.04     0.09     0.23 1.00
## cor(drift_Intercept,drift_condition1)      -0.93      0.03    -0.97    -0.85 1.00
## cor(ndt_Intercept,ndt_condition1)          -0.20      0.26    -0.68     0.32 1.00
##                                         Bulk_ESS Tail_ESS
## sd(drift_Intercept)                          791     1372
## sd(drift_condition1)                        1012     1848
## sd(bound_Intercept)                          570     1163
## sd(ndt_Intercept)                           1276     2181
## sd(ndt_condition1)                          1084     1185
## cor(drift_Intercept,drift_condition1)       1300     2008
## cor(ndt_Intercept,ndt_condition1)           1362     1536
##
## Regression Coefficients:
##                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## drift_Intercept       1.18      0.07     1.04     1.32 1.01      329      751
## bound_Intercept       0.62      0.04     0.55     0.70 1.01      259      644
## ndt_Intercept        -0.87      0.05    -0.97    -0.79 1.00      755     1494
## drift_condition1     -0.84      0.06    -0.96    -0.71 1.00      371      866
## ndt_condition1        0.26      0.04     0.19     0.33 1.00     1996     2816
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
ezdm_hypothesis <- c(
  driftFX = "drift_condition1 = 0",
  ndtFX = "ndt_condition1 = 0"
)

ezdm_hypothesis_results <- brms::hypothesis(
  ezdm_fit,
  ezdm_hypothesis
)

plot(ezdm_hypothesis_results)
```




```r
1/abs(ezdm_hypothesis_results$hypothesis$Evid.Ratio)
```

```
## [1] 6.205778e+24 3.646612e+16
```